

K.R. MANGALAM UNIVERSITY
B.TECH COMPUTER SCIENCE (2ND YEAR)
END-TERM EXAMINATION QUESTION BANK:
VOLUME 2

Unit 1: Differential Equations

LINEAR DE

[10 Marks]

Q: Solve: $dy/dx + y \cot(x) = 2x \operatorname{cosec}(x)$.

Step 1: Standard form $dy/dx + P y = Q$, where $P = \cot(x)$ and $Q = 2x \operatorname{cosec}(x)$.

Step 2: Integrating Factor (IF) = $e^{\int \cot(x) dx} = e^{\ln(\sin(x))} = \sin(x)$.

Step 3: Solution is $y * IF = \int Q * IF dx + C$.

Step 4: $y \sin(x) = \int (2x \operatorname{cosec}(x)) * \sin(x) dx = \int 2x dx$.

Step 5: $y \sin(x) = x^2 + C$.

Answer: $y = x^2 \operatorname{cosec}(x) + C \operatorname{cosec}(x)$

Unit 1

EXACT DE

[10 Marks]

Q: Solve: $(2x + 3y)dx + (3x + 2y)dy = 0$.

Step 1: Let $M = 2x + 3y$, $N = 3x + 2y$.

Step 2: $\partial M / \partial y = 3$, $\partial N / \partial x = 3$. Since they are equal, the DE is exact.

Step 3: $\int M dx = \int (2x + 3y) dx = x^2 + 3xy$ (treating y as constant).

Step 4: $\int N dy$ (ignoring terms with x) = $\int 2y dy = y^2$.

Step 5: Combine the integrated terms and equate to C .

Answer: $x^2 + 3xy + y^2 = C$

Q: Solve the boundary value problem: $y'' + 9y = 0$, with $y(0) = 1$ and $y(\pi/6) = 2$.

Step 1: Auxiliary equation $m^2 + 9 = 0 \Rightarrow m = \pm 3i$.

Step 2: General solution: $y(x) = C_1 \cos(3x) + C_2 \sin(3x)$.

Step 3: Apply $y(0) = 1$: $1 = C_1 \cos(0) + C_2 \sin(0) \Rightarrow C_1 = 1$.

Step 4: Apply $y(\pi/6) = 2$: $2 = (1)\cos(\pi/2) + C_2 \sin(\pi/2) \Rightarrow 2 = 0 + C_2(1) \Rightarrow C_2 = 2$.

Step 5: Substitute C_1 and C_2 back into the general solution.

Answer: $y(x) = \cos(3x) + 2\sin(3x)$

Unit 2: Numerical Methods

Q: Find a real root of $x e^x = 2$ correct to 2 decimal places using $x_0 = 0.8$.

Step 1: Let $f(x) = x e^x - 2$. Then $f'(x) = e^x + x e^x = e^x(1 + x)$.

Step 2: Iteration 1: $f(0.8) = 0.8(2.2255) - 2 = -0.2195$. $f'(0.8) = 2.2255(1.8) = 4.0059$.

Step 3: $x_1 = x_0 - f(x_0)/f'(x_0) = 0.8 - (-0.2195)/4.0059 = 0.8 + 0.0548 = 0.8548$.

Step 4: Iteration 2: $f(0.8548) = 0.003$, $f'(0.8548) = 4.36$.

Step 5: $x_2 = 0.8548 - 0.003/4.36 = 0.8541$.

Answer: $x \approx 0.85$

Unit 2

Q: Solve the system using 1 iteration of Gauss-Seidel: $5x - y = 9$, $-x + 5y = 11$ (Initial guess $x=0$, $y=0$).

Step 1: Check diagonal dominance ($5 > |-1|$, $5 > |-1|$). It is strictly diagonally dominant.

Step 2: Rearrange equations: $x = (9 + y) / 5$; $y = (11 + x) / 5$.

Step 3: Iteration 1 for x: Use $y_0=0$. $x_1 = (9 + 0) / 5 = 1.8$.

Step 4: Iteration 1 for y: Use newest x ($x_1=1.8$). $y_1 = (11 + 1.8) / 5 = 12.8 / 5 = 2.56$.

Answer: $x_1 = 1.8$, $y_1 = 2.56$

LU DECOMPOSITION

[10 Marks]

Q: Find L and U for matrix $A = \begin{bmatrix} 3 & 1 \\ 6 & 4 \end{bmatrix}$.

Step 1: Let $A = LU$. $L = \begin{bmatrix} 1 & 0 \\ L_{21} & 1 \end{bmatrix}$, $U = \begin{bmatrix} U_{11} & U_{12} \\ 0 & U_{22} \end{bmatrix}$.

Step 2: $LU = \begin{bmatrix} U_{11} & U_{12} \\ L_{21} U_{11} & L_{21} U_{12} + U_{22} \end{bmatrix}$. Equate to A.

Step 3: $U_{11} = 3$, $U_{12} = 1$.

Step 4: $L_{21} * U_{11} = 6 \Rightarrow L_{21} * 3 = 6 \Rightarrow L_{21} = 2$.

Step 5: $L_{21} * U_{12} + U_{22} = 4 \Rightarrow 2 * 1 + U_{22} = 4 \Rightarrow 2 + U_{22} = 4 \Rightarrow U_{22} = 2$.

Answer: $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$

Unit 3: Random Variables

POISSON DISTRIBUTION

[10 Marks]

Q: If the probability of producing a defective bolt is 0.02, find the probability that a box of 100 bolts contains exactly 3 defectives.

Step 1: Identify $n = 100$, $p = 0.02$. This is a binomial case that can be approximated by Poisson.

Step 2: Calculate Poisson parameter $\lambda = n * p = 100 * 0.02 = 2$.

Step 3: Formula $P(X=x) = (e^{-\lambda} * \lambda^x) / x!$.

Step 4: $P(X=3) = (e^{-2} * 2^3) / 3! = (0.1353 * 8) / 6 = 1.0824 / 6$.

Step 5: $P(X=3) = 0.1804$.

Answer: 0.1804 (or 18.04%)

Unit 3

CONTINUOUS RV

[10 Marks]

Q: Given PDF $f(x) = kx$ for $0 \leq x \leq 2$, find the constant k and the Mean $E[X]$.

Step 1: The total integral of $f(x)$ from 0 to 2 must equal 1.

Step 2: $\int (kx)dx = [k x^2 / 2]$ from 0 to 2 = $k(4)/2 = 2k$. So, $2k = 1 \Rightarrow k = 1/2$.

Step 3: Mean $E[X] = \int x \cdot f(x) dx$ from 0 to 2.

Step 4: $E[X] = \int x \cdot (x/2) dx = \int (x^2 / 2) dx = [x^3 / 6]$ from 0 to 2.

Step 5: $E[X] = 8/6 = 4/3$.

Answer: $k = 1/2$, Mean = $4/3$

Unit 4: Optimization

LAGRANGE MULTIPLIER

[10 Marks]

Q: Minimize $f(x,y) = x^2 + y^2$ subject to the constraint $2x + 3y = 13$.

Step 1: Let $g(x,y) = 2x + 3y - 13$. Construct $L = (x^2 + y^2) - \lambda(2x + 3y - 13)$.

Step 2: $\partial L / \partial x = 2x - 2\lambda = 0 \Rightarrow x = \lambda$.

Step 3: $\partial L / \partial y = 2y - 3\lambda = 0 \Rightarrow y = (3/2)\lambda$.

Step 4: Substitute x and y into the constraint: $2(\lambda) + 3(1.5\lambda) = 13$.

Step 5: $2\lambda + 4.5\lambda = 13 \Rightarrow 6.5\lambda = 13 \Rightarrow \lambda = 2$.

Step 6: $x = 2$, $y = 3$. Minimum value = $2^2 + 3^2 = 4 + 9 = 13$.

Answer: Min value is 13 at (2,3)

Unit 4

LINEAR REGRESSION

[10 Marks]

Q: Fit a straight line $y = a + bx$ to points: (1,3), (2,5), (3,7).

Step 1: Prepare sums: $n=3$, $\Sigma x=6$, $\Sigma y=15$, $\Sigma x^2 = 1+4+9 = 14$, $\Sigma xy = 3+10+21 = 34$.

Step 2: Normal Eq 1: $\Sigma y = na + b\Sigma x \Rightarrow 15 = 3a + 6b \Rightarrow 5 = a + 2b$ (dividing by 3).

Step 3: Normal Eq 2: $\Sigma xy = a\Sigma x + b\Sigma x^2 \Rightarrow 34 = 6a + 14b \Rightarrow 17 = 3a + 7b$ (dividing by 2).

Step 4: Multiply Eq 1 by 3: $15 = 3a + 6b$. Subtract from Eq 2: $(17 - 15) = (3a - 3a) + (7b - 6b)$.

Step 5: $2 = b$. Substitute $b=2$ into $5 = a + 2b \Rightarrow 5 = a + 4 \Rightarrow a = 1$.

Answer: $y = 1 + 2x$